

RN00192

for S32K312 HSE Firmware 0.2.55.0

Rev. 1.1 — 7 February 2025

Release notes
COMPANY CONFIDENTIAL

Document information

Information	Content
Keywords	HSE, S32K312, HSE Firmware, RN00192
Abstract	This release note is specific to the HSE Firmware for the S32K312 device. It describes the changes with each update of the HSE S32K312 Firmware along with known issues for each corresponding version.



1 Getting Started

IMPORTANT NOTES:

This is the Standard Package variant of the HSE Firmware for S32K312 device.

1.1 Package content

This package contains the NXP S32K312 HSE Firmware 0.2.55.0 (Service Release):

- FULL MEM HSE Firmware: encrypted binary
- AB SWAP HSE Firmware: encrypted binary
- HSE Firmware interface files
- HSE Service API RM
- HSE_FW_S32K312_0.2.55.0_ReleaseNotes.pdf – this file
- The license.txt file and the `uninstall.exe` utility for removing the HSE Firmware binaries

Note:

Demo Application is provided separately and contains details on how to provision HSE Firmware on new device from factory and demonstrates common use cases of its security features. For this release, RM can be provided On-Demand. The Demo Application can be accessed through the section Automotive SW - S32K3 - HSE Firmware Demo App on Flexera.

SBAF Encrypted Binary has been uploaded on Flexera which can be accessed through the section Automotive SW - S32K3 - Secure BAF (SBAF). Using this binary, the SBAF can be updated on a device using the service ID HSE_SRV_ID_SBAF_UPDATE.

One can access via NXP DocStore (<https://www.docstore.nxp.com>) the following associated documentation:

- HSE Firmware Reference Manual VERSION 2.4

1.2 Installation

Follow the install steps in the demo application.

2 Release Details

This is the HSE Firmware 0.2.55.0 Standard FW Service Release.

This release is a Service Release and has RFP (RTM) quality in terms of testing and quality documentation. Service Release contains all the features and is fully tested on Customer Engineering Samples or Qual Intend Samples. RFP (RTM) release is delivered with complete software documentation, and quality package.

The provided example code shows how to setup and use the HSE Firmware and to perform basic cryptographic operations (refer to the documentation that comes with the demo application). The examples will show how to:

- Boot the demo-application (secure mode)
- Install the Firmware
- Load the key(s)
- Perform cryptographic operations

This release was developed and tested using:

- Chip : E5 - P32K312NHVPBS
- Mini-Module : XS32K3X2CVB-Q172

Implemented Errata:

N/A

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP:

- S32K312

2.2 Device Bricking scenario for HSE Firmware

- No bricking scenario identified in *CUST_DEL* lifecycle.

2.3 Security Aspects

Current release of HSE Firmware implements partial counter-measures against logical attacks (e.g. input parameter checking, address ranges). The code contained in this release was not subject to penetration testing / vulnerability attacks verification.

3 Changes List

3.1 Changes in 0.2.55.0

Added

- Key verify service extended to support asymmetric keys.
- Check for data flash write operation and acquire lock before accessing KHTT.
- TRNG initialization from UTEST.

Updated

- Downgraded User access to `HSE_SRV_ID_ACTIVATE_PASSIVE_BLOCK` from Super User to Normal User.
- HSE debug codes have been extended.
- To keep the compatibility, the support for SHA384 and SHA512 for key verify has been added back.
- Added details in the interface of `HSE_SECURE_RECOVERY_CONFIG_ATTR_ID`.
- Use the same item size for both Config Data and KHTT.
- In ABSwap reduce HSE init time after Partition swap and reset by comparing Hash of Active and Passive firmware.
- Updated the CRC verification process so that it does not fail when reading KHTT from NVM during loading NVM data in HSE Init.

Issues Fixed

- RESET SOC sanction not working in post boot.
- Key Drive Verification for `HSE_HASH_ALGO_SHA2_384` and `HSE_HASH_ALGO_SHA2_512` Expectation Error shows General Error.
- Mismatch between `pSmrDest` and `pSmrSize` alignment check condition and interface comment.
- SMR update as it did not check if `pPassReset` was in range and allowed it if outside SMR region.
- State of the Sensor was changing from ARM/DISARM state to Unused State using `HSE_SENSOR_DISARMING_ON_STARTUP_ATTR_ID` and gives no error.
- Zero length input for ECDSA sign that caused fault attack is fixed.
- Prevent race condition in WDT idle/running state change.
- ECDSA Digital Signature generation and verification service when `bInputIsHashed` is true.
- Invalid redeclaration of type name.
- Alignment of the SHE KEY to the correct group in Key catalog.
- `GET_ATTR` service to allow the retrieval of the `HSE_SECURE_RECOVERY_CONFIG_ATTR_ID`.

Removed

- Consistency check when importing a ECC/RSA key pair.

3.2 Changes in 0.2.40.1 (Hotfix)

Fixed

- Fixed a rare scenario where HSE went into shutdown with error code 0x33AA0001. The issue is related to HSE going to Shutdown when service calls are initiated at 5 second periods.

3.3 Changes in 0.2.40.0

Added

- Added new service (`HSE_SRV_ID_ERASE_FW`) to remove HSE Firmware and sys image from the device (only allowed in CUST_DEL lifecycle). This will enable reconfiguration of the device.

Updated

- Improved interface comments must Erase NVM Data and updated placement order of service IDs
- Added and updated description of HSE GPR status bits related to sys image sector getting erased by HSE FW when corruption happens.

Removed

- Disabled following services to increase available code flash for HSE FW:
 - Burmester Desmedt (`HSE_SPT_BURMESTER_DESMEDT`)
 - Extend Key Catalogs (`HSE_SPT_EXTEND_KEY_CATALOG`)
 - Classic DH (`HSE_SPT_CLASSIC_DH`)
 - App Specific Data Attribute (`HSE_SPT_APP_SPECIFIC_DATA_ATTR`)
 - CMAC with Counter (`HSE_SPT_CMAC_WITH_COUNTER`)
- Removed the support macro `HSE_SPT_ECC_COMPRESSED_KEYS`. The feature is still present in HSE FW.

3.4 Changes in 0.2.32.0

Added

- This release candidate contains ECC error handling change request for both Secure-BAF and HSE Firmware. In full memory configuration, to get the ECC error handling feature in Secure-BAF, the user must update the Secure-BAF using the service ID `HSE_SRV_ID_SBAF_UPDATE`.
- Added the sys image integrity feature in `HSE_SRV_ID_FW_INTEGRITY_CHECK` service.

Updated

- Increment counter is made as heavy job internally instead of fast job.
- First service depending on monotonic counters will take more time. After that they will be executed in normal way. This is done to save boot time and handle possible ECC errors in monotonic counter flash area.
- Removed `HSE_SPT_NXP_ROM_PUB_KEYS` and added `HSE_SPT_NXP_ROM_RSA_PUB_KEYS` and `HSE_SPT_NXP_ROM_ECC_PUB_KEYS`
- Renamed FW Type macro in `hse_target.h` (`STANDARD` -> `HSE_STANDARD_FW`)
- Added and updated description of HSE GPR status bits related to sys image sector getting erased by HSE FW when corruption happens because of ECC error.
- While configuring the ADKP via attribute `HSE_APP_DEBUG_KEY_ATTR_ID`, HSE Firmware returns error `HSE_SRV_RSP_INVALID_PARAM`, if filled with all 0x00 and 0xFF.

Issues Fixed

- HSE Firmware always returned `HSE_SRV_RSP_OK` for HMAC using SHA384 verification and HMAC using SHA512 verification.

3.5 Changes in 0.2.31.0

Added

- Add Support for SHA3 functionality

3.6 Changes in 0.2.12.0

Added

- Added the partial AB swap feature. Applicable only for S32K358 devices.
- JDC Disable support added for S32K358 devices.

Issues Fixed

- `smrEntryInstallStatus` is not getting updated even after getting `ERC_NO_ERR` for SMR entry erase service.

3.7 Changes in 0.2.7.0

Added

- Support for KM catalog extend service.
- Support for ECC ROM keys for all devices involving trust center. Please contact NXP for more details on this.
- Support to store app data inside HSE FW.
- Support for SHA 384 and SHA 512 in ANSI 9.63 KDF.
- Provided the flexibility in firmware and SBAF to not use JDC for debug authorization on request. This is applicable only for S32K396 device family.
- Support of Deletion of SMR Entry & CR Entry.

Updated

- Updated the window time for time slice mechanism. Fast jobs will have 4 times more execution time as compared to heavy jobs.

Issues Fixed

- Master secret computation fails using TLS 1.2 PRF for DHE PSK.

3.8 Changes in 0.2.8.0

N/A

3.9 Changes in 0.2.6.0

Updated

- Firmware Update service for same Firmware Type in FULL MEM configuration will be allowed only if code size and sys image size is equal or higher than the current Firmware code size and sys-image size installed in the system.
- In the case of streaming mode, if firmware input length given has already exceeded the expected pink image length, then error will be thrown, and firmware update process will be terminated.
- In case there is firmware update request from standard to custom firmware for FULL MEM configuration, and there is change in sys image or code size, then also firmware update request will be allowed only in one shot mode.
- In case of change in sys image or code size change in new HSE Firmware image, then Firmware Update will be allowed only in one shot mode.

Issues Fixed

- Monotonic Counter Increment Service when run in parallel to Asymmetric Services (RSA & ECC) causes failures in both services and causes HSE FW to go into shutdown.
- Monotonic Counter Increment Service when ran in parallel to HASH/HMAC services causes failures in both services and causes HSE FW to go into shutdown.
- Fast CMAC without scatter gather given after Fast CMAC with scatter gather always fails with CAAM hardware error.
- HSE Firmware Update does not complete when XRDC is enabled by the host application. This issue is not applicable for S32K344 devices.

3.10 Changes in 0.2.1.0

Added

- Added the feature of calculating and verifying the GMAC using random IV in IVT, Base secure boot and secure recovery image
- Support for compressed ECC keys - `hseEccKeyFormat_t`
- Key verify service - `hseKeyVerifySrv_t` – generates a hash/MAC over a symmetric key from HSE key store
- AAD support for encrypted SMR - `hseSmrDecrypt_t`
- Enable the support of NXP ROM keys to allow the import of first user root key in secure manner. Please contact NXP support team for more details.
- Added support for Publish Sys Image and Attribute to allow user to program multiple keys in the flash once. This will help in reduction in time to store multiple keys in flash
- NVM Erase to be done in case junk data is present in the device and inform the status to the user.
- Added the support of software based HMAC-384 and HMAC-512 functionality.
- AES block mode mask as part of the AES key info. It restricts the usage of an AES key only to those specified by `hseAesBlockModeMask_t` member. If set to 0 then no restrictions apply
- Added the support of TLS RSA master secret key feature
- Enabled the support of Diffie-Hellman key exchange support.
- Added the support of address translation logic in SMR verification service for AB swap configuration. `hseSmrVerificationOptions_t`, `hseSmrVerifySrv_t`
- Added the support of scatter gather entries in service “HSE_SRV_ID_CMAC_WITH_COUNTER” - `hseCmacWithCounterSrv_t`
- The key derive copy restrictions on starting offset and number of bytes extracted - `hseKeyDeriveCopyKeySrv_t`

Updated

- Improve the flash programming check logic in interface level function. HSE Firmware will return an error at the time of accepting the request only when flash programming is in progress on the same code flash block where HSE Firmware is executing.
- Updated the comments in interface files.
- Minimum MAC length to 8 bytes for MAC service
- Include the MU instance in streaming context
- Policy on encrypted keys import/export. If encrypted, it must also be authenticated
- Minimum MAC length to 16 bytes for plain SMR initial authentication proof, authenticated key import/export and system authorization services
- RNG re-initialization is done for one time when first heavy job is requested after HSE Firmware initialization is completed.

Updated the ITCM and DTCM memory address range for output address for S32K314 device.

Issues Fixed

- HSE status and error bits are not updated on the application side (MU.FSR and MU.GSR) until `HSE_STATUS_INIT_OK` is set
- Flash Read while Write Issue is observed in Firmware if Flash Operation is ongoing on the device.

Removed

- XRDC configuration in SBAF to improve the boot time.
- Specific fatal error events for tamper violations - `hseError_t`
- SHA1 support for key derivation services

- HMAC and CMAC support as PRF for NXP key derivation scheme

3.11 Changes in 0.1.2.1 (hotfix)

Issues Fixed

- HSE goes into the exception due to RWW(read while write error) while handling watchdog ISR, if the Host is performing the flash operation (programming/ erase) on the same block on which HSE Firmware is executing. (ASHF-4611)

3.12 Changes in 0.1.2.0

Added

- Service `HSE_SRV_ID_IMPORT_EXPORT_STREAM_CTX`. This service is used for Import/Export streaming context for the symmetric operation.
- Enhanced the synchronization logic for the Flash read and write operation between the Host application and the HSE core. For every flash block, Status bit in the GPR register are provided which indicates to the Host application that HSE core is performing read or write operation on a specific block. For more details (Refer to HSE Firmware reference manual V1.2)
- Added the `HSE_RAM_PUB_KEY_IMPORT_POLICY_ATTR_ID` attribute which allows the importing of public keys in RAM without authentication.
- Added the SMR rollback protection feature.
- Added the support of glitch filter enablement for the Active Tamper.

Updated

- Interface comments.
- Key Import:
- Only authentication is required for importing the RAM provision keys. These keys can be used to import only RAM keys.
- NVM provisioning keys can be imported/updated without authentication only having SuperUser rights.
- Key Export:
- NVM keys cannot be exported using RAM provision keys.
- NVM and RAM symmetric keys can be exported only encrypted and authentication is optional.

Issues Fixed

- User Keys in the NVM are getting erased after giving a reset. (ASHF-4435)
- TLS 1.2 KDF output is not correct when secret key length is greater than the block size of the Hash Algo. (ASHF-4303)

3.13 Change logs in 0.1.1.0

Added

- Service `HSE_SRV_ID_IMPORT_EXPORT_STREAM_CTX`. This service is used for Import/Export streaming context for the symmetric operation.
- Enhanced the synchronization logic for the Flash read and write operation between the Application and the HSE core. For every flash block, Status bit in GPR are provided which indicates to host application that HSE core is performing read or write operation on a specific block. For more details (Refer to HSE Firmware reference manual V1.1)

Updated

- Interface comments.

3.14 Change logs in 0.0.12.0

Added

- Added the `HSE_RAM_PUB_KEY_IMPORT_POLICY_ATTR_ID` attribute which allows the importing the public keys in RAM without authentication.
- Added the SMR rollback protection feature.
- Added the support of glitch filter enablement in case of active tamper and updated the description for the same in interface file.

Updated

- Interface comments and enhancement. Some items have been moved to new headers (to decongest some headers).
- Key Import:
 - Only authentication is required for importing the RAM provision keys. These keys can be used to import only RAM keys.
 - NVM provisioning keys can be imported/updated without authentication only having SuperUser rights.
- Key Export:
 - NVM keys cannot be exported using RAM provision keys.
 - NVM and RAM symmetric keys can be exported only encrypted and authentication is optional.

3.15 Change logs in 0.0.10.0

Added

- Streaming support in HSE Firmware update service.
- The size of `HSE_MAX_RAM_STORE_SIZE` increased to 6144 bytes from 4096 bytes.
- Service `HSE_SRV_ID_FW_INTEGRITY_CHECK`. This service checks the integrity of active HSE Firmware in code flash area and backup image in data flash area.
- Service `HSE_SRV_ID_BURMESTER_DESMEDT`. This service implements the ECC Burmester-Desmedt Protocol to calculate the shared secret between multiple nodes i.e., more than 2.
- Service `HSE_SRV_ID_SBAF_UPDATE`. This service is used to update the SBAF on the device whenever the Firmware image is not compatible with SBAF present in the system.
- Service `HSE_SRV_ID_CMACE_WITH_COUNTER`. This service is used to calculate and verify the CMACE by appending the monotonic counter in input message.
- Handling of tamper violations: clock monitoring (CMU) and physical tamper.
- Service `HSE_SRV_ID_ON_DEMAND_CORE_RESET`. This service allows configuring CR entries and their associated SMR to be processed at run-time, on-demand
- Event `HSE_WA_SMR_PERIODIC_CHECK_FAILED`. This event signals the host that a periodic check SMR failed (the verification failed).
- Event `HSE_WA_DATA_FLASH_INTEGRITY_FAIL`. This event signals the host that valid backup of HSE Firmware is not present in data flash.
- Register HSE GPR for tamper status. This register is updated by HSE when a tamper is configured. It can be read by the host to check what tampers are configured.
- Prevention of the accesses to shared memory of other cores via HSE (see `hseAttrAllMuMemRegions_t` attribute).
- Service versioning using a byte from the service ID to encode the version for each service.
- Scatter-Gather support for RSA and ECDSA signature (refer to `hseSignSrv_t`)
- Added support for Adkp Provisioning via Key Handle.
- Added `HSE_CORE_RESET_RELEASE_ATTR_ID` attribute to configure the core release-from-reset strategy:
 - all-at-once (cores are release all at once after all boot SMRs (Secure Memory Region) are verified).

- one-by-one (cores are release one by one as soon as the boot SMR(s) verification passed for that core)
- New HSE attributes (see *hseAttrId_t*):
 - *HSE_PHYSICAL_TAMPER_ATTR_ID* – Enablement of physical tamper.
 - *HSE_MEM_REGIONS_PROTECT_ATTR_ID* – Configure memory regions accessible through each MU.
 - *HSE_SECURE_RECOVERY_CONFIG_ATTR_ID* – Enablement of the secure recovery mechanism.
 - *HSE_CORE_RESET_RELEASE_ATTR_ID* – Release core from reset.

Updated

- Interface comments.
- The minimum value of Rollover Protection bits in monotonic counter is set to 32 bits and the size of Rollover Protection bits must be multiple of 8 bits.
- The configuration of monotonic counter can be done using only Super User rights.
- SHE keys catalog formatting must be configured for *HSE_KEY_OWNER_ANY* group owner.
- *HSE_SRV_ID_GET_RANDOM_NUM* macro value updated.
- Firmware is not functional if *HSE_FIRC_DIVIDER_CONFIG_ATTR_ID* is received with request to divide by 16.
- *HSE_ENABLE_BOOT_AUTH_ATTR_ID* attribute is not enabled if GMAC tag is not programmed in any of the IVT locations on the device.
- The *HSE_SRV_ID_ERASE_HSE_NVM_DATA* service does not require resetting the device to re-use the firmware services. This service is allowed only in CUST-DEL lifecycle.
- Secure Memory Regions (SMR) (see *hseSmrEntry_t*):
 - Added support for encrypted SMR. The SMRs can be encrypted using GCM or CTR.
 - Removed the SMR verification method field (*hseSmrVerifMethod_t*).
 - The SMR periodic tick has been updated from *10ms* to *100ms* (at 120 MHz frequency).
 - SMR version number added to provide the anti-rollback protection for the image against attacks during update.
- Core Reset (CR) (see *hseCrEntry_t*):
 - Included *PRE-BOOT*, *ALT_PRE_BOOT* and *POST-BOOT* SMR bit map in Core Reset entries.
 - Included *hseCrStartOption_t* option: auto-start (automatically release the core from reset at start-up) or on-demand (the core boot is triggered on demand by another application core).
 - For unsecure boot (*BOOT_SEQ=0*), the SMRs are not loaded at boot time. Application can use the *hseSmrVerifySrv_t* service to load and verify the SMR (for validation purpose).
 - Updated the SHE Secure Boot used with SMR entry#0 (see *hseSmrEntryInstallSrv_t* comments).
- Updated *hseSmrVerifySrv_t* service to: Verify any SMR using service ID *HSE_SRV_ID_SMR_VERIFY*.
- Fast CMAC:
 - Use input and tag length in bits.
 - Included *HSE_FAST_CMAC_MIN_TAG_BIT_LEN_ATTR_ID* attribute to configure the minimum tag bit-length that can be used for Fast CMAC verify / generate.
- HSE errors reported to HOST are divided into warnings and errors.
- Error events reported by HSE (see *hseError_t*). To clear the errors, the host must read the MUB_GSR register and write back the register value (W1C)
- Application header for Basic Secure Boot: “tag address” and “key type” fields were removed; “core ID” field not used; core booted is specified by *BOOT_TARGET* in IVT; the tag must be placed after the application code (see *hseAppHeader_t*, *hseBootDataImageSignSrv_t*)
- Import/Export key to support GCM and CCM: AAD, Tag and IV should be specified in the AEAD scheme, and the keyInfo (key proprieties) must be within AAD (see *hseImportKeySrv_t*, *hseExportKeySrv_t*)
- HMAC to support key sizes greater than hash block size (updated *hseMacSrv_t* service)

- Erase service to delete the SHE keys only if system authorization was performed beforehand using MASTER_ECU key. Other keys (non-SHE keys) can be erased if the authorization operation was performed using any key type, including MASTER_ECU key (see *hseEraseKeySrv_t*)
- Load ECC service to save the ECC user-curve domain parameters in SYS-IMG (which needs to be published). The host needs Super User rights to be able to load an ECC user-defined curve. The loaded ECC user-defined curves must have the private key size equal to or greater than 192 bits
- SYS Authorization feature (see *hseSysAuthorizationReqSrv_t*) to perform the authorization using the *SHE MASTER_ECU_KEY*. SHE keys can be erased only if the host is authorized with MASTER_ECU_KEY.
- The host must call the *HSE_SRV_ID_ACTIVATE_PASSIVE_BLOCK* service to load new HSE Firmware after successful execution of the HSE Firmware update service in case of AB swap configuration.
- The SYS-IMG configured for 0.0.8.3 Firmware release is not compatible with this release. The Host must reconfigure the SYS-IMG.
- The size of HSE Firmware increased from 80KB to 128KB, the implications of this change would be:
- AB swap firmware cannot be installed through older version of Secure BAF (Boot Assist Flash).
- The Host must free up 40KB from DATA Flash out of 128KB. Hence, DATA Flash range that can be used by the Host is [0x10000000 – 0x10015FFF].
- The Host must free up 168KB from CODE Flash memory,
- Available CODE Flash range in FULL MEM configuration [0x400000 – 0x7D3FFF].
- Available CODE Flash range in AB SWAP configuration
ACTIVE BLOCK: [0x400000 – 0x5D3FFF]
PASSIVE BLOCK: [0x600000 – 0x7D3FFF]
- **Removed**
 - All TDES support.
 - IV Length parameter from symmetric cipher (see *hseSymCipherSrv_t*)
 - *HSE_KDF_SP800_108_FEEDBACK* and *HSE_KDF_SP800_108_PIPELINE* KDF SP800 modes. Only Counter Mode remained supported.
 - The restriction on allowing an SMR to be updated, in advanced life cycles (OEM_PROD, IN_FIELD), only if it is already verified successfully. (ASHF-3433)

4 Known Issues

- N/A

5 List of Limitations

- The HSE Firmware is operational when HSE_CLK is between 24MHz to 120MHz.
- User should restart the device once Firmware Update is completed in FULL_MEM configuration.
- In GCM operations, it is recommended to use IV values of exactly 12 bytes. For any IV size different from 12 bytes, the authentication tag generated by the GCM encryption operation might not be correct. In GCM decryption operations, the authentication check might fail.

6 HSE FW and SBAF Compatibility

Table 1. Compatibility for HSE FW use with SBAF updates

SI No	NPI	Derivatives	STD HSE FW Version (This Release)	HSE FW			SBAF		
				Previous/Last STD HSE FW Version	Is this release backward compatible?	Remarks	Previous/Last SBAF Version	Is this release compatible?	Remarks
1	S32K311	K310 K311							
2	S32K312	K312	0.2.55.0	0.2.40.1	Yes	-	00 0D 00 00 00 0F 00 06 (0.15.0)	Yes	Older versions of SBAF are not compatible with this release.
3	S32K342	K342 K341 K322	0.2.55.0	0.2.40.1	Yes	-	00 0D 00 00 00 0F 00 06 (0.15.0)	Yes	Older versions of SBAF are not compatible with this release.
4	S32K344	K344 K324 K314	0.2.55.0	0.2.40.0	Yes	-	00 05 00 00 00 0F 00 06 (0.15.0)	Yes	Older versions of SBAF are not compatible with this release.
5	S32K396	K396 K366 K364							
6	S32K358	K358 K348 K338 K328							
7	S32K388	K388							
8	S32K389	K389							

7 HSE Firmware Version and S32K3XX Device Compatibility

The Table below explains the HSE Firmware version number compatibility with the S32K3 family devices.

Table 2. HSE FW Version Compatibility with S32K3 Devices

S.NO	HSE Firmware version number [<Configtype>_ <socTypeID>_ <fwTypeld>_ <majorVersion>_ <minorVersion>_ <patchVersion>]	Remarks	Applicable HSE Firmware reference manuals	Release Date
1	s32k312_hse_fw_0.13.0_2.55.0_pb250129.bin (FullMem) s32k312_hse_fw_1.13.0_2.55.0_pb250129.bin (AB Swap)	HSE Firmware Service Release compatible with S32K312 Devices. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual – V2.4	7 Feb 2025
2	s32k312_hse_fw_0.13.0_2.40.1_pb240711.bin.pink (Full Mem) s32k312_hse_fw_1.13.0_2.40.1_pb240711.bin.pink (AB Swap)	HSE Firmware Hotfix release compatible with S32K312 Devices. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual – V2.3	12 Jul 2024
3	s32k312_hse_fw_0.13.0_2.40.0_pb230730.bin (FullMem) s32k312_hse_fw_0.13.0_2.40.0_pb230730.bin (AB Swap)	HSE Firmware RTM release compatible with S32K312 Devices. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual – V2.2	2 Aug 2023
4	s32k312_hse_fw_0.13.0_2.32.0_pb230620.bin (FullMem) s32k312_hse_fw_1.13.0_2.32.0_pb230620.bin (AB Swap)	HSE Firmware Standard Code Drop release compatible with S32K312 Devices. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual – V2.1	20 Jun 2023
5	s32k3x2_hse_fw_0.13.0_2.6.0_pb221129.bin (FullMem) s32k3x2_hse_fw_1.13.0_2.6.0_pb221129.bin (AB Swap)	HSE Firmware Standard RTM release compatible with S32K312 Device. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual – V2.0	9 Dec 2022
6	s32k3x2_hse_fw_0.13.0_1.2.1_pb220205.bin (FullMem) s32k3x2_hse_fw_1.13.0_1.2.1_pb220205.bin (AB Swap)	HSE Firmware Standard hotfix release compatible with S32K312 Device. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual - V1.2	07 Feb 2022
7	s32k3x2_hse_fw_0.13.0_1.2.0_pb211228.bin (FullMem) s32k3x2_hse_fw_1.13.0_1.2.0_pb211228.bin (AB Swap)	HSE Firmware Standard BETA release compatible with S32K312 Device. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual - V1.2	18 Jan 2022

Table 2. HSE FW Version Compatibility with S32K3 Devices...continued

8	s32k3x2_hse_fw_0.13.0_0.11.0_pb210726.bin (FullMem) s32k3x2_hse_fw_1.13.0_0.11.0_pb210726.bin (AB Swap)	HSE Firmware Standard EAR release compatible with S32K312 Device. This release supports both Full Mem/AB Swap configuration.	HSE-B Firmware Reference Manual - V0 - Draft M	16 Aug 2021
---	--	---	---	----------------

8 List of Services Available

All available HSE features/services are also listed in the *hse_b_config.h* file (from HSE Interface). All other features not listed in the table below (or enabled in *hse_b_config.h* file) are **NOT supported**. All available services applicable for both configuration device.

Table 3. Available HSE Features/Services

Service Class	HSE Service ID	Description/Notes
Administrative	HSE_SRV_ID_SET_ATTR	Set an HSE attribute. Attributes related to UTEST memory can be written only once (e.g. Debug Key) or can only be advanced (e.g. Life cycle). Care must be taken.
	HSE_SRV_ID_GET_ATTR	Get an HSE attribute.
	HSE_SRV_ID_CANCEL	Cancel a one-pass or streaming service on a specific channel. An HSE service request can be cancelled if it is in the processing queue and NOT passed to the hardware to be executed.
	HSE_SRV_ID_FIRMWARE_UPDATE	HSE Firmware update This service covers the following things: 1. Firmware update in FullMem configuration for Standard Firmware with one-pass/streaming mode 2. Firmware update in ABSwap configuration for Standard Firmware with one-pass/streaming mode
	HSE_SRV_ID_SBAF_UPDATE	Secure-BAF update in FullMem configuration for Standard Firmware with one-pass/streaming mode
	HSE_SRV_ID_SYS_AUTH_REQ	SYS Authorization request used to request the Super User rights (CUST/OEM).
	HSE_SRV_ID_SYS_AUTH_RESP	SYS Authorization response (response to SYS Authorization Request)
	HSE_SRV_ID_BOOT_DATA_IMAGE_SIGN	Generate the signature on Boot Header & XRDC image using ADKP
	HSE_SRV_ID_BOOT_DATA_IMAGE_VERIFY	Verify the signature on Boot Header & XRDC image using ADKP
	HSE_SRV_ID_IMPORT_EXPORT_STREAM_CTX	Import and Export service for the crypto streaming context.
Key Management	HSE_SRV_ID_ERASE_HSE_NVM_DATA	Erase HSE Data Flash only in customer delivery lifecycle
	HSE_SRV_ID_ERASE_FW	Erase the HSE Firmware in code flash, back up area and sys image
	HSE_SRV_ID_FW_INTEGRITY_CHECK	Triggers the integrity check of HSE Firmware code flash and data flash area.
	HSE_SRV_ID_PUBLISH_NVM_KEYSTORE_RAM_TO_FLASH	Application requests the firmware to write the NVM keys from RAM mirrored keystore into the data flash. This service has no parameters.
	HSE_SRV_ID_LOAD_ECC_CURVE	Load the domain parameters for a Weierstrass ECC curve. This service can be used to support additional Weierstrass ECC curves (Not supported by default). The loaded ECC curve domain parameters are persistent.
	HSE_SRV_ID_FORMAT_KEY_CATALOGS	Format key application key catalogs (RAM & NVM).

Table 3. Available HSE Features/Services...continued

Service Class	HSE Service ID	Description/Notes
	HSE_SRV_ID_ERASE_KEY	Erase NVM/RAM key(s). Erase key service depends on authorization rights. One or multiple keys can be erased. Selected key group can also be erased
	HSE_SRV_ID_GET_KEY_INFO	Get key proprieties (flags).
	HSE_SRV_ID_IMPORT_KEY	Import a key. Uses all algorithms supported by HSE firmware: * Plain form or AES / RSA encrypted. * MAC authenticated (refer to supported MAC algorithms) or RSA / ECDSA signed. * Import key restrictions depends on sys authorization rights. The restrictions are described by the service in the interface.
	HSE_SRV_ID_EXPORT_KEY	Export a key. Uses all algorithms supported by HSE firmware: * Plain form (only public keys) or AES / RSA encrypted. * MAC authenticated (refer to supported MAC algorithms) or RSA / ECDSA signed. * Export key restrictions depends on authorization rights. The restrictions are described by the service in the interface.
	HSE_SRV_ID_KEY_GENERATE	Request to generate a symmetric/asymmetric key. * Random symmetric key generation * RSA and ECC key pair generation
	HSE_SRV_ID_DH_COMPUTE_SHARED_SECRET	ECC Diffie-Hellman Compute Key (shared secret): * SEC curves: SECP256R1, SECP384R1, SECP521R1 * Brainpool curves: BRAINPOOLP256R1, BRAINPOOLP320R1, BRAINPOOLP384R1, BRAINPOOLP512R1 * Montgomery curve: CURVE25519 * 1 user-defined ECC curves (see Load ECC curve service)
	HSE_SRV_ID_KEY_DERIVE	Perform a key derivation function: * NXP Generic KDF, SP800_56C One Step, SP800_56C Two Step, ANSI_X963, SP800_108 (Only Counter Mode), ISO/IEC 18033 KDF1, ISO/IEC 18033 KDF2, PBKDF2HMAC, HKDF, TLS12PRF
	HSE_SRV_ID_KEY_DERIVE_COPY	Extract a key from the derived key material to a key slot.
	HSE_SRV_ID_KEY_VERIFY	Computes a hash/MAC over a symmetric key inside HSE key store.
	HSE_SRV_ID_SHE_LOAD_KEY	Load a SHE key using the SHE memory update protocol.
	HSE_SRV_ID_SHE_LOAD_PLAIN_KEY	Loads SHE specification-based RAM key as plain text.
	HSE_SRV_ID_SHE_EXPORT_RAM_KEY	Exports RAM key as per SHE specification.
	HSE_SRV_ID_SHE_GET_ID	GETUID as per SHE specification.
	HSE_SRV_ID_SHE_BOOT_OK	The command is used to mark successful boot verification for later stages than CMD_SECURE_BOOT. For more details, see SHE specification

Table 3. Available HSE Features/Services...continued

Service Class	HSE Service ID	Description/Notes
	HSE_SRV_ID_SHE_BOOT_FAILURE	The command will impose the same sanctions as if CMD_SECURE_BOOT would detect a failure but can be used during later stages of the boot process. For more details, see SHE specification.
ROM Keys	N/A	Support for ROM keys (only AES keys).
Cryptographic	HSE_SRV_ID_HASH	Hash service (one-pass and streaming): * SHA1 * SHA-224, SHA-256, SHA-384, SHA-512, SHA-512/224, SHA-512/256 * Miyaguchi-Preneel compression function (SHE specification support)
	HSE_SRV_ID_MAC	Request to generate/verify a Message Authentication Code (MAC): * AES-CMAC, AES-GMAC * HMAC_(SHA1, SHA_224, SHA_256, SHA_384, SHA_512)
	HSE_SRV_ID_FAST_CMIC	Low latency, high performance CMIC generate/verify request
	HSE_SRV_ID_SYM_CIPHER	Symmetric encryption/decryption (one-pass and streaming): * AES-128/-192/-256: ECB, CBC, CTR, OFB, CFB
	HSE_SRV_ID_AEAD	AEAD encryption/decryption: * AES-CCM-128/-192/-256 (one-pass, no streaming support) * AES-GCM-128/-192/-256 (one-pass and streaming)
	HSE_SRV_ID_SIGN	Request a Digital Signature Generation/Verification (one-pass and streaming): * RSASAA_PSS (1024, 2048, 3072, 4096) * RSASAA_PKCS1-v1_5(1024, 2048, 3072, 4096) * ECDSA (all supported ECC curves) * EDDSA (for ED25519 curve)
	HSE_SRV_ID_RSA_CIPHER	RSA encryption/decryption: * RSAES-PKCS1-v1_5 (1024, 2048, 3072, 4096) * RSAES-OEAP (1024, 2048, 3072, 4096)
RNG	HSE_SRV_ID_GET_RANDOM_NUM	Get a random number. AIS31 and FIPS 140-2 compliant
Counters	HSE_SRV_ID_INCREMENT_COUNTER	Incrementing Non-volatile counter.
	HSE_SRV_ID_READ_COUNTER	Read Non-volatile counter.
	HSE_SRV_ID_CONFIG_COUNTER	Configure the secure counter.
Advance Secure Booting (SMR/CR)	HSE_SRV_ID_SMR_ENTRY_INSTALL	Install a Secure Memory Region (SMR) table entry.
	HSE_SRV_ID_SMR_VERIFY	Verify (on demand) a Secure Memory Region (SMR) table entry.
	HSE_SRV_ID_CORE_RESET_ENTRY_INSTALL	Install a Core Reset (CR) table entry.
	HSE_SRV_ID_ON_DEMAND_CORE_RESET	On-demand release a core from a reset after loading and verification

Table 3. Available HSE Features/Services...continued

Service Class	HSE Service ID	Description/Notes
	HSE_SRV_ID_SMR_ENTRY_ERASE	Erase an SMR Entry.
	HSE_SRV_ID_CORE_RESET_ENTRY_ERASE	Erase a CR Entry.

The service below is applicable only for **AB SWAP** configuration device.

Table 4. Services for AB SWAP Configuration Device

Service Class	HSE Service ID	Description/Notes
Administrative	HSE_SRV_ID_ACTIVATE_PASSIVE_BLOCK	Switch to passive area. Using this service application can switch to passive area.

9 Acronyms

Table 5. Acronyms

Acronym	Meaning / Description
ADKP	Application Debug Key / Password; one-time programmable parameter
AES	Advanced Encryption Standard; a cipher primitive
BCW	Boot Configuration Word
BAF	Boot Assist Flash
CUST_DEL	Device Life Cycle Customer Delivery
ECC	Error Correction Code
EL	Edge Lock
FA	Device Life Cycle Failure Analysis
FDK	Firmware Delivery Key
FW	Firmware
FXOSC	Fast External Crystal Oscillator
HBM	Host Boot Manager
HDR	Header
HMAC	Hash-based Message Authentication Code
HSE	Hardware Security Engine
ISO	International Organization for Standardization
IN_FIELD	Device Life Cycle is IN_FIELD
IVT	Image Vector Table
LC	Device Life Cycle; used to limit by design the configuration and debug / test possibilities of the device for in-field usage
MAC	Message Authentication Code
MCU_PROD	Device Life Cycle MCU Production
MDA	Master Domain Assignment
MRC	Memory Region Controller
MRGD	Memory Region Descriptor
MU	Messaging Unit
N/A	Not Applicable
NSK	NXP Signing Key
OEM_PROD	Device Life Cycle OEM Production
OTA	Over The Air
OTP	One Time Programmable
PDAC	Peripheral Domain Access Control
RNG	Random Number Generator
RSA	A public-key cryptosystem named after the inventors Mr. Rivest, Mr. Shamir and Mr. Adleman
SBAF/Secure BAF	Secure Boot Assist flash. Boot component in HSE.

Table 5. Acronyms...continued

Acronym	Meaning / Description
VT	Verification Tag

10 Revision History

Table 6. Revision history

Document ID	Release Date	Description
RN00192v1.1	7 February 2025	Release notes for S32K312 HSE Firmware 0.2.55.0 Service Release
RN00192v.1	11 July 2024	Release notes for S32K312 HSE Firmware 0.2.40.1 Hotfix

Legal information

Definitions

Draft — A draft status on a document indicates that the content is still under internal review and subject to formal approval, which may result in modifications or additions. NXP Semiconductors does not give any representations or warranties as to the accuracy or completeness of information included in a draft version of a document and shall have no liability for the consequences of use of such information.

Disclaimers

Limited warranty and liability — Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms and conditions of commercial sale of NXP Semiconductors.

Right to make changes — NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Applications — Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Terms and conditions of commercial sale — NXP Semiconductors products are sold subject to the general terms and conditions of commercial sale, as published at <https://www.nxp.com/profile/terms>, unless otherwise agreed in a valid written individual agreement. In case an individual agreement is concluded only the terms and conditions of the respective agreement shall apply. NXP Semiconductors hereby expressly objects to applying the customer's general terms and conditions with regard to the purchase of NXP Semiconductors products by customer.

Suitability for use in automotive applications — This NXP product has been qualified for use in automotive applications. If this product is used by customer in the development of, or for incorporation into, products or services (a) used in safety critical applications or (b) in which failure could lead to death, personal injury, or severe physical or environmental damage (such products and services hereinafter referred to as "Critical Applications"), then customer makes the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, safety, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP. As such, customer assumes all risk related to use of any products in Critical Applications and NXP and its suppliers shall not be liable for any such use by customer. Accordingly, customer will indemnify and hold NXP harmless from any claims, liabilities, damages and associated costs and expenses (including attorneys' fees) that NXP may incur related to customer's incorporation of any product in a Critical Application.

Export control — This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

HTML publications — An HTML version, if available, of this document is provided as a courtesy. Definitive information is contained in the applicable document in PDF format. If there is a discrepancy between the HTML document and the PDF document, the PDF document has priority.

Translations — A non-English (translated) version of a document, including the legal information in that document, is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Security — Customer understands that all NXP products may be subject to unidentified vulnerabilities or may support established security standards or specifications with known limitations. Customer is responsible for the design and operation of its applications and products throughout their lifecycles to reduce the effect of these vulnerabilities on customer's applications and products. Customer's responsibility also extends to other open and/or proprietary technologies supported by NXP products for use in customer's applications. NXP accepts no liability for any vulnerability. Customer should regularly check security updates from NXP and follow up appropriately.

Customer shall select products with security features that best meet rules, regulations, and standards of the intended application and make the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP.

NXP has a Product Security Incident Response Team (PSIRT) (reachable at PSIRT@nxp.com) that manages the investigation, reporting, and solution release to security vulnerabilities of NXP products.

NXP B.V. — NXP B.V. is not an operating company and it does not distribute or sell products.

Trademarks

Notice: All referenced brands, product names, service names, and trademarks are the property of their respective owners.

NXP — wordmark and logo are trademarks of NXP B.V.

Tables

Tab. 1.	Compatibility for HSE FW use with SBAF updates	14	Tab. 4.	Services for AB SWAP Configuration Device	20
Tab. 2.	HSE FW Version Compatibility with S32K3 Devices	15	Tab. 5.	Acronyms	21
Tab. 3.	Available HSE Features/Services	17	Tab. 6.	Revision history	23

Contents

1	Getting Started	2
1.1	Package content	2
1.2	Installation	2
2	Release Details	3
2.1	Supported Derivatives	3
2.2	Device Bricking scenario for HSE Firmware	3
2.3	Security Aspects	3
3	Changes List	4
3.1	Changes in 0.2.55.0	4
3.2	Changes in 0.2.40.1 (Hotfix)	4
3.3	Changes in 0.2.40.0	4
3.4	Changes in 0.2.32.0	5
3.5	Changes in 0.2.31.0	5
3.6	Changes in 0.2.12.0	5
3.7	Changes in 0.2.7.0	6
3.8	Changes in 0.2.8.0	6
3.9	Changes in 0.2.6.0	6
3.10	Changes in 0.2.1.0	7
3.11	Changes in 0.1.2.1 (hotfix)	8
3.12	Changes in 0.1.2.0	8
3.13	Change logs in 0.1.1.0	8
3.14	Change logs in 0.0.12.0	9
3.15	Change logs in 0.0.10.0	9
4	Known Issues	12
5	List of Limitations	13
6	HSE FW and SBAF Compatibility	14
7	HSE Firmware Version and S32K3XX Device Compatibility	15
8	List of Services Available	17
9	Acronyms	21
10	Revision History	23
	Legal information	24

Please be aware that important notices concerning this document and the product(s) described herein, have been included in section 'Legal information'.